
Indian Railways Digital Twin Project Plan

Dataset, Simulation, and Agentic Optimization Roadmap

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Project	Indian Railways Digital Twin Dataset, Simulation, and Agentic Optimization Platform

1. Executive Objective

Build a production-minded digital twin for Indian Railways that combines infrastructure, timetable, live movement, incident, weather, workshop, and operational context into a single simulation-ready dataset. The platform should support delay diagnosis, replay, scenario simulation, dispatch optimization, and human-supervised agent recommendations.

The immediate priority is not to automate railway control. The first priority is to create a trusted data foundation and simulation environment where every recommendation is traceable, measurable, and auditable before any operational rollout.

2. Product Vision

The final system should act as an operations intelligence layer over the rail network:

- Show the current state of trains, stations, track segments, workshops, incidents, and constraints.
- Reconstruct what happened historically at minute-level resolution.
- Simulate future timetable and disruption scenarios before they happen.
- Recommend dispatch actions such as hold, priority push, platform change, route swap, and workshop-aware rerouting.
- Compare policy options by delay, conflict count, safety constraints, passenger impact, and operational feasibility.
- Keep a human dispatcher in control with full action audit history.

3. Current Foundation

The repository already contains the core dataset and bootstrap foundation:

- Canonical SQLite schema for stations, tracks, trains, timetables, runs, movement events, incidents, weather context, simulation batches, state snapshots, KPIs, and dispatch actions.
- Synthetic and national-scale seed generators.
- CSV ingestion for zone packages and external curated tables.
- Connector ingestion skeleton for NTES, IRCTC, and public timetable style feeds.
- Open-Meteo weather enrichment.
- Incident-to-run linkage.
- Simulation and baseline dispatch optimizer.
- Agentic headway and disruption search loop.
- Feature export for ML training.
- Dataset bundle export with manifest.
- Dynamic web dashboard for bundle inspection and operational KPIs.

4. Core Approach

Dataset First

All operational intelligence depends on trustworthy data. The project will treat the canonical dataset as the first product. Every source must map into explicit tables with lineage, timestamps, reliability score, and validation checks.

Connector First, Synthetic as Fallback

Synthetic data is useful for development, but production planning must move toward connector-backed live and historical ingestion. Target sources:

- NTES or approved train running status and timetable feeds.
- IRCTC or approved incident/disruption feeds.
- Public railway timetable datasets where licensing allows operational analysis.
- Internal or partner CSV packages for stations, tracks, workshops, and maintenance windows.
- Weather and event context for external disruption modeling.

Simulation Before Automation

The system should first recommend actions inside a replay/simulation environment. Actions should be promoted only after repeated evaluation against historical scenarios and safety constraints.

Human-in-the-Loop Agents

Agents should optimize candidate policies, not directly control trains. The agent layer should produce ranked recommendations with reasons, confidence, and expected KPI impact.

5. Target Architecture

1. Source Layer

- NTES timetable and train-running connector.
- IRCTC incident/disruption connector.
- Public timetable connector.
- Weather/event connector.
- CSV/partner ingestion lane for infrastructure and workshop data.

2. Canonical Data Layer

- Source registry.
- Station and track graph.
- Train and timetable catalog.
- Daily train run table.
- Movement event timeline.
- Incident and external context tables.

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- Train-incident link table.

3. Processing Layer

- Validation and quality checks.
- Deduplication and timestamp normalization.
- Incident-run reconciliation.
- Headway and occupancy feature generation.
- Multi-day bootstrap and bundle export.

4. Simulation Layer

- Rule-based baseline simulator.
- Headway and conflict detection.
- Delay propagation model.
- Workshop and maintenance constraint modeling.
- Scenario replay and what-if evaluation.

5. Optimization Layer

- Dispatch action candidate generator.
- Agentic policy search.
- KPI scoring and safety guardrails.
- Human approval workflow.

6. Experience Layer

- Dynamic operations dashboard.
- Dataset health dashboard.
- Simulation comparison view.
- Action audit and recommendation history.
- Exportable reports for review.

6. Roadmap

Phase 0: Stabilize Foundation

Duration: 1 week

Goal: Make the current project easy to run, inspect, and demo.

Deliverables:

- One-command local bootstrap.
- Dashboard launch instructions.
- Clean dataset bundle outputs.

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- Project-plan PDF and stakeholder handoff pack.
 - Known-good sample database and manifest.

Exit targets:

- Local bootstrap completes without manual edits.
- Dashboard loads bundle and project manifest.
- Quality checks pass for seeded profile.

Phase 1: Real-World Data Ingestion

Duration: 2 to 3 weeks

Goal: Replace synthetic-only runs with provider-approved live and historical inputs.

Deliverables:

- Live connector configuration for approved timetable feeds.
- Live incident connector configuration.
- Public timetable adapter.
- Provider payload mapping tests.
- Source lineage and reliability scoring.
- Dry-run and production-run command profiles.

Exit targets:

- At least one live timetable feed ingests into canonical tables.
- At least one incident/disruption source ingests into canonical tables.
- Every ingested row is traceable to source, batch, and service date.
- Failed or partial source pulls are visible in quality reports.

Phase 2: Network and Asset Completeness

Duration: 3 to 4 weeks

Goal: Build a richer rail graph with operational constraints.

Deliverables:

- Station, segment, gauge, electrification, speed, platform, and single-line data coverage.
- Workshop and maintenance capacity model.
- Yard, siding, workshop access, and main-line classification.
- Zone and division enrichment.
- First complete pilot corridor or zone.

Exit targets:

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- Pilot zone coverage: 95 percent station completeness.
 - Pilot track graph connectivity: 98 percent valid station-segment links.
 - Workshop capacity captured for major maintenance nodes in pilot area.
 - Track segment metadata available for speed, line type, and electrification.

Phase 3: Historical Reconstruction

Duration: 4 to 6 weeks

Goal: Build replayable historical state timelines.

Deliverables:

- Daily run reconstruction.
- Minute-level movement timeline.
- Delay attribution pipeline.
- Incident-to-run linkage confidence.
- Baseline historical KPI reports.

Exit targets:

- 6 to 12 months of pilot-zone history, subject to data availability.
- 90 percent or higher run-timetable matching for ingested timetable sources.
- 80 percent or higher incident attribution coverage where incidents overlap active runs.
- Reproducible daily bundles for all reconstructed days.

Phase 4: Simulation and Scenario Engine

Duration: 4 to 6 weeks

Goal: Simulate normal and disrupted operations.

Deliverables:

- Baseline delay propagation model.
- Platform and headway conflict model.
- Disruption scenario runner.
- KPI comparison reports.
- Scenario bundle export.

Exit targets:

- Replay simulation produces stable KPIs across repeated runs.
- Scenario comparison shows action impact on delay, conflicts, and on-time rate.
- Every suggested action has source state, reason, and computed impact.

Phase 5: Agentic Optimization

Duration: 4 to 8 weeks

Goal: Move from static rules to ranked, policy-aware recommendations.

Deliverables:

- Agent loop for headway and dispatch threshold search.
- Constraint engine for safety, capacity, and operational feasibility.
- Multi-objective scoring.
- Action ranking with confidence and reason.
- Human approval state machine.

Exit targets:

- Agent recommendations improve simulated KPI baseline on replay scenarios.
- No recommendation violates configured safety constraints.
- All recommendations include audit fields: actor, scope, reason, before state, after state, and expected KPI effect.

Phase 6: Operational Dashboard and Pilot Rollout

Duration: 4 to 6 weeks

Goal: Put the system in front of operators and analysts for review.

Deliverables:

- Live dashboard refresh from latest bundle/API.
- Corridor-level map and topology view.
- Incident command timeline.
- Simulation comparison workspace.
- Operator feedback capture.
- Deployment checklist.

Exit targets:

- Analyst can inspect current data health, latest runs, incidents, and recommended actions.
- Operator can approve/reject recommendations in a non-control pilot.
- Feedback loop captures why actions were accepted or rejected.

7. Target KPIs

Data Quality Targets

- Source lineage completeness: 100 percent of rows with source or batch traceability.

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- Required field validity: 99 percent for core station, track, train, and timetable fields.
 - Timestamp validity: 99 percent for movement events and incidents.
 - Duplicate event rate: below 1 percent after reconciliation.
 - Timetable-to-run matching: 90 percent or higher in pilot zone.

Operational Modeling Targets

- Simulation replay stability: same scenario produces same KPI outputs.
- Delay attribution coverage: 80 percent or higher for disruptions with matching train activity.
- Conflict detection coverage: all headway and platform conflicts in simulated state snapshots.
- Feature export completeness: 95 percent or higher for modeled runs.

Optimization Targets

- Average delay reduction in replay scenarios: 5 to 12 percent initial target.
- Conflict count reduction: 10 percent or higher in controlled scenarios.
- On-time percentage improvement: 3 to 8 percentage points in pilot simulations.
- Unsafe recommendation rate: 0 percent under configured constraints.
- Human-review traceability: 100 percent of generated recommendations.

Product Targets

- Local bootstrap time: under 10 minutes for pilot dataset.
- Dashboard data load: under 5 seconds for pilot bundle.
- Daily pipeline success rate: 95 percent or higher after connector stabilization.
- Bundle generation: one complete manifest per service date.

8. Governance and Safety

The project must avoid pretending to be a train-control system before it is validated. Recommended governance:

- Use production data only through approved provider terms and secure credentials.
- Keep all agent output as recommendations until formal operational approval.
- Store every recommendation with reason, actor, source data, before state, after state, and KPI estimate.
- Add hard constraints before any optimization output is trusted.
- Separate simulation environment from operational systems.
- Require human approval for all operational actions during pilot.
- Maintain audit logs for all data ingest, transformations, simulations, and recommendations.

9. Risks and Mitigations

Feed Access and Licensing

Risk: NTES, IRCTC, or public timetable feeds may be rate-limited, incomplete, or restricted.

Mitigation: Use approved connectors, dry-runs, source registry metadata, and fallback CSV import lanes.

Incomplete Track Graph

Risk: Missing or outdated track/yard/workshop data can distort simulation.

Mitigation: Start with one pilot corridor, validate graph connectivity, and expand by zone.

Historical Data Drift

Risk: Historical timetables and live events may not align cleanly.

Mitigation: Add reconciliation confidence, preserve raw source references, and flag unresolved mismatches.

Unsafe Optimization

Risk: An agent may suggest operationally invalid actions.

Mitigation: Gate all recommendations through hard constraints, audit logs, and human approval.

Overfitting to Synthetic Data

Risk: Synthetic results may look better than real-world performance.

Mitigation: Treat synthetic as development only and measure final progress on live/historical replay data.

10. Next Execution Backlog

1. Fix local dashboard serving flow and add optional auto-load from latest bundle.
2. Add provider-specific mapping tests for live timetable and incident payloads.
3. Build a pilot corridor configuration file with stations, segments, workshops, and target sources.
4. Add connector run logs and source freshness table.
5. Add quality report export to PDF/HTML.
6. Add a scenario comparison report for baseline versus optimized simulation batch.
7. Add map/topology view to dashboard.
8. Add recommendation approval states: proposed, reviewed, accepted, rejected, expired.
9. Add action-impact estimate fields to dispatch actions.
10. Prepare stakeholder demo bundle with one live-seed run when credentials are available.

11. Immediate 30-Day Targets

- Week 1: Polish project handoff, dashboard, bootstrap command, and known-good bundle.

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- Week 2: Connect first approved timetable feed and run dry-run plus canonical ingest.
 - Week 3: Connect incident feed, reconcile incidents to runs, and export quality report.
 - Week 4: Run pilot corridor simulation and compare baseline versus optimized actions.

12. Definition of Done for Pilot

The pilot is ready when:

- A selected corridor or zone can be ingested from approved real-world feeds.
- The system produces daily canonical bundles with source lineage.
- Quality checks pass or produce clear actionable failures.
- A historical or live-seed replay produces simulation KPIs.
- The optimizer generates ranked recommendations with reasons.
- The dashboard shows health, KPIs, incidents, actions, and rollout timeline.
- Human reviewers can inspect recommendations without relying on code or SQL.

13. Recommended Operating Principle

Build confidence in this order:

1. Data correctness.
2. Replay correctness.
3. Simulation usefulness.
4. Recommendation quality.
5. Human approval workflow.
6. Operational integration.

That order keeps the project grounded. It also makes it possible to measure whether the twin is actually reducing delay risk instead of only producing visually impressive outputs.